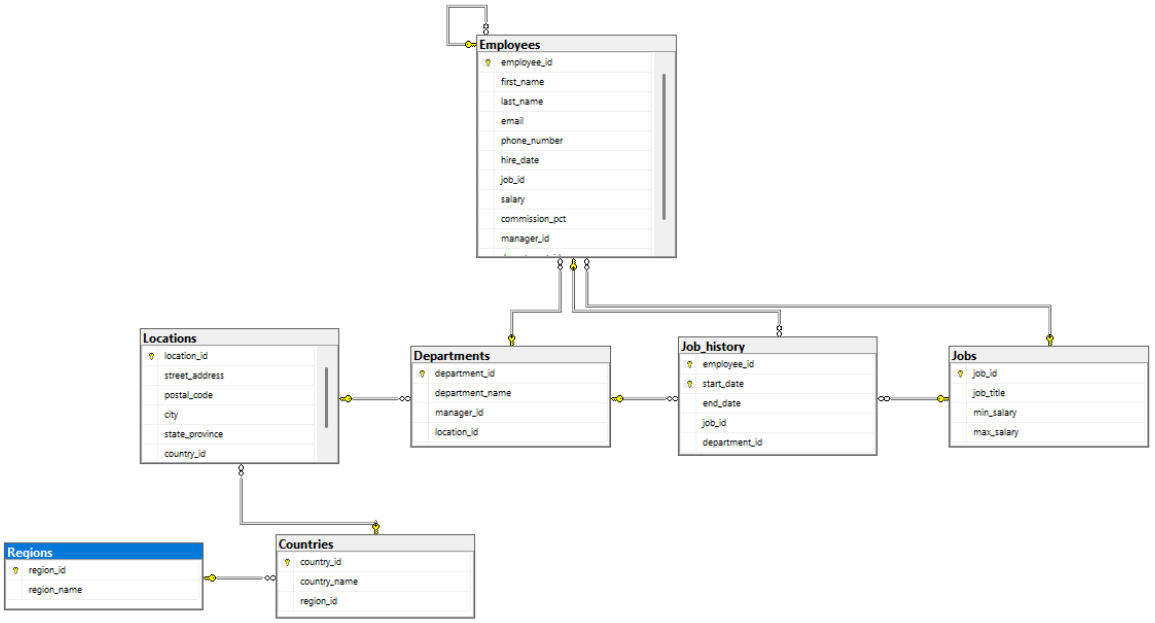
Name: A.Sanjay kumar
Email: attellisanjay29@gmail.com

Day 1

Use hrdb;



Question 1
Create a table named Regions with region_id as primary key and region_name.

```
CREATE TABLE Regions (
   region_id INT PRIMARY KEY,
   region_name VARCHAR(50)
);
```

Question 2
Insert three records into the Regions table.

```
INSERT INTO Regions VALUES
(1, 'Asia'),
(2, 'Europe'),
(3, 'America');
```

Results | Messages

	region_id	region_name
1	1	Asia
2	2	Europe
3	3	America

Question 3
Create a table named Jobs with a check constraint to ensure minimum salary is less than or equal to maximum salary

```
CREATE TABLE Jobs (
   job_id INT PRIMARY KEY,
   job_title VARCHAR(20) NOT NULL,
   min_salary DECIMAL(10,2),
   max_salary DECIMAL(10,2),
   CONSTRAINT chk_salary CHECK (min_salary <= max_salary)
);
```

Question 4
Insert sample data into the Jobs table.

```
INSERT INTO Jobs VALUES
(101, 'Manager', 50000, 100000),
(102, 'Developer', 30000, 80000),
(103, 'Tester', 25000, 60000);
```

Results | Messages

	job_id	job_title	min_salary	max_salary
1	101	Manager	50000.00	100000.00
2	102	Developer	30000.00	80000.00
3	103	Tester	25000.00	60000.00

Question 5
Create a Countries table and link it with Regions using a foreign key.

```
CREATE TABLE Countries (
   country_id CHAR(5) PRIMARY KEY,
   country_name VARCHAR(50),
   region_id INT NOT NULL,
   FOREIGN KEY (region_id) REFERENCES Regions(region_id)
);
```

Question 6
Insert data into the Countries table.

```
INSERT INTO Countries VALUES
('IN', 'India', 1),
('UK', 'United Kingdom', 2),
('US', 'United States', 3);
```

Results | Messages

	country_id	country_name	region_id
1	IN	India	1
2	UK	United Kingdom	2
3	US	United States	3

Question 7
Create a Locations table with a foreign key reference to Countries.

```
CREATE TABLE Locations (
    location_id INT PRIMARY KEY,
    street_address VARCHAR(100),
    postal_code VARCHAR(20),
    city VARCHAR(15) NOT NULL,
    state_province VARCHAR(50),
    country_id CHAR(5),
    FOREIGN KEY (country_id) REFERENCES Countries(country_id)
);
```

Question 8
Insert records into the Locations table.

```
INSERT INTO Locations VALUES
(1, 'MG Road', '500001', 'Hyderabad', 'Telangana', 'IN'),
(2, 'Oxford Street', 'W1D', 'London', 'England', 'UK'),
(3, '5th Avenue', '10001', 'New York', 'NY', 'US');
```

Results | Messages

	location_id	street_address	postal_code	city	state_province	country_id
1	1	MG Road	500001	Hyderabad	Telangana	IN
2	2	Oxford Street	W1D	London	England	UK
3	3	5th Avenue	10001	New York	NY	US

Question 9
Create a Departments table linked to Locations.

```
CREATE TABLE Departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(50) NOT NULL,
    manager_id INT NULL,
    location_id INT,
    FOREIGN KEY (location_id) REFERENCES Locations(location_id)
);
```

Question 10
Insert data into the Departments table.

```
INSERT INTO Departments VALUES
(10, 'HR', NULL, 1),
(20, 'IT', NULL, 2),
(30, 'Finance', NULL, 3);
```

Results | Messages

	department_id	department_name	manager_id	location_id
1	10	HR	NULL	1
2	20	IT	NULL	2
3	30	Finance	NULL	3

Question 11
Create an Employees table with foreign keys referencing Jobs and Departments.

```
CREATE TABLE Employees (
   employee_id INT PRIMARY KEY,
   first_name VARCHAR(100),
   last_name VARCHAR(100),
   email VARCHAR(50),
   phone_number VARCHAR(20),
   hire_date DATE,
   job_id INT,
   salary DECIMAL(10,2),
   commission_pct DECIMAL(10,2),
   manager_id INT NULL,
   department_id INT NULL,
   FOREIGN KEY (job_id) REFERENCES Jobs(job_id),
   FOREIGN KEY (department_id) REFERENCES Departments(department_id),
   FOREIGN KEY (manager_id) REFERENCES Employees(employee_id)
);
```

Question 12
Insert sample employee data.

```
INSERT INTO Employees VALUES
(1, 'Sanjay', 'Kumar', 'sanjay@mail.com', '900000001', '2024-01-10', 102, 60000, NULL,
NULL, 20),
(2, 'Ravi', 'Sharma', 'ravi@mail.com', '900000002', '2023-03-15', 101, 90000, NULL, 1, 10),
(3, 'Neha', 'Singh', 'neha@mail.com', '900000003', '2022-06-20', 103, 40000, NULL, 2, 30);
```

Results | Messages

	employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id
1	1	Sanjay	Kumar	sanjay@mail.com	900000001	2024-01-10	102	60000.00	NULL	NULL	20
2	2	Ravi	Sharma	ravi@mail.com	900000002	2023-03-15	101	90000.00	NULL	1	10
3	3	Neha	Singh	neha@mail.com	900000003	2022-06-20	103	40000.00	NULL	2	30

Question 13
Create a Job_history table to store past job details of employees.

```
CREATE TABLE Job_history (
    employee_id INT,
    start_date DATE,
    end_date DATE,
    job_id INT,
    department_id INT,
    PRIMARY KEY (employee_id, start_date),
    FOREIGN KEY (employee_id) REFERENCES Employees(employee_id),
    FOREIGN KEY (job_id) REFERENCES Jobs(job_id),
    FOREIGN KEY (department_id) REFERENCES Departments(department_id)
);
```

Question 14
Insert sample data into Job_history.

```
INSERT INTO Job_history VALUES
(1, '2022-01-01', '2023-12-31', 103, 30),
(2, '2021-05-01', '2022-12-31', 102, 20);
```

Results | Messages

	employee_id	start_date	end_date	job_id	department_id
1	1	2022-01-01	2023-12-31	103	30
2	2	2021-05-01	2022-12-31	102	20